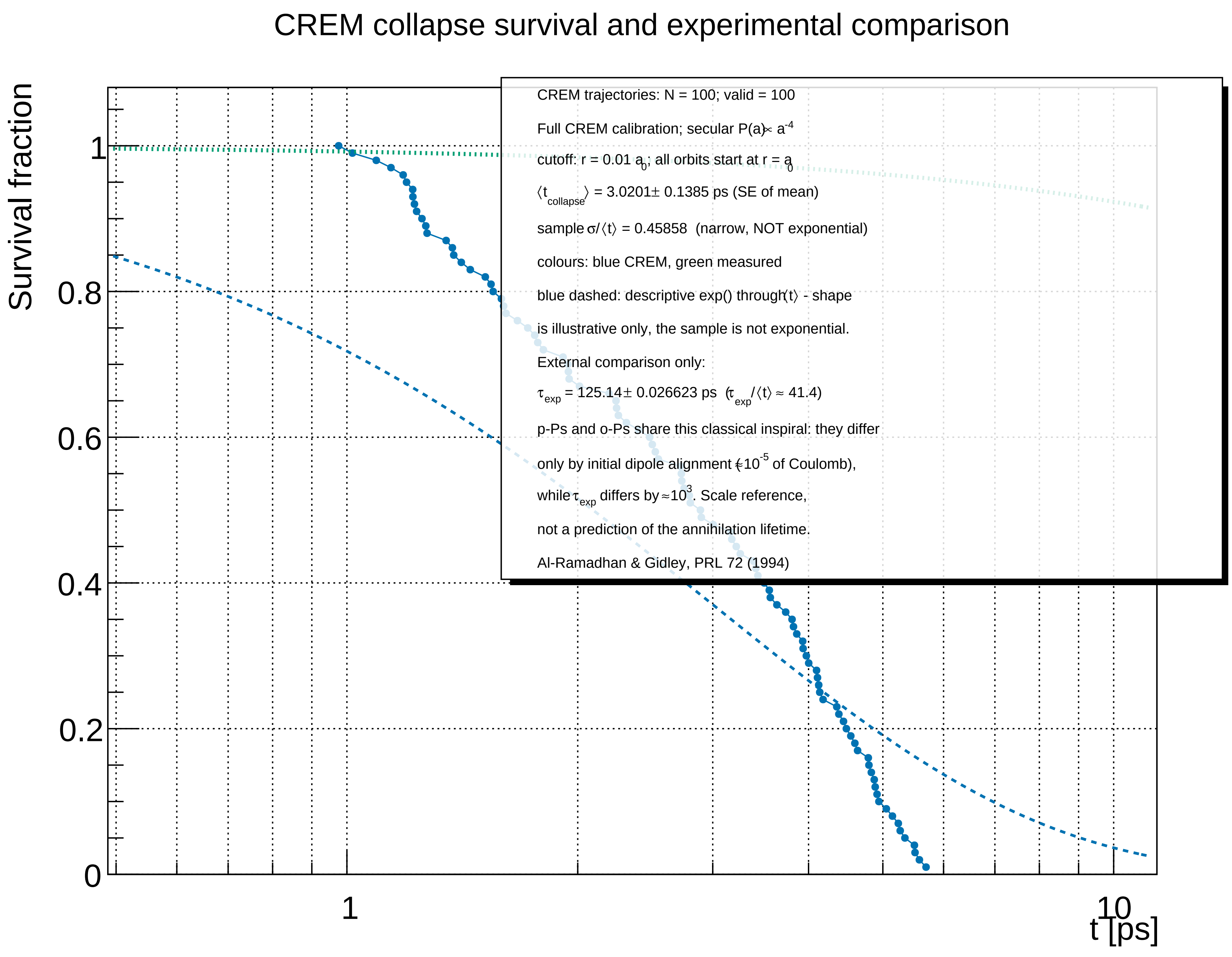
CREM collapse survival and experimental comparison
Survival fraction
t [ps]
0
0.2
0.4
0.6
0.8
1
1
10
CREM trajectories: N = 100; valid = 100
Full CREM calibration; secular P(a) ∝ a^-4
cutoff: r = 0.01 a0; all orbits start at r = a0
⟨t collapse⟩ = 3.0201± 0.1385 ps (SE of mean)
sample σ/⟨t⟩ = 0.45858 (narrow, NOT exponential)
colours: blue CREM, green measured
blue dashed: descriptive exp() through ⟨t⟩ - shape
is illustrative only, the sample is not exponential.
External comparison only:
τ exp = 125.14± 0.026623 ps (τ exp/⟨t⟩ ≈ 41.4)
p-Ps and o-Ps share this classical inspiral: they differ
only by initial dipole alignment (≤10^-5 of Coulomb),
while τ exp differs by ≈10^3. Scale reference,
not a prediction of the annihilation lifetime.
Al-Ramadhan & Gidley, PRL 72 (1994)